Protein Secondary Structure Prediction Report

By

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Abstract

This project applies machine learning to predict alpha-helices in protein secondary structures. Using biochemical scales, amino acid sequence data, and structural annotations, three machine learning models were implemented: Random Forest, Logistic Regression, and K-Nearest Neighbors (KNN). After hyperparameter tuning, the Random Forest model achieved the highest accuracy of 78.4% on the test set. The study underscores the potential of computational biology in understanding protein structures, laying the foundation for future research in bioinformatics.

Introduction

Proteins are macromolecules essential to life, and their functionality depends heavily on their structures. Secondary structures, such as alpha-helices, are local folding patterns within polypeptide chains that influence a protein's overall structure and function. Predicting these structures computationally is critical for applications in drug discovery, synthetic biology, and molecular modeling.

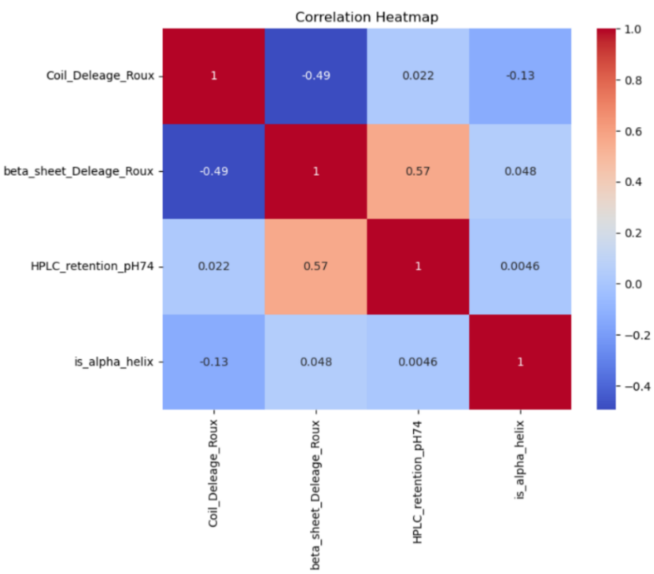
In this study, machine learning techniques were applied to classify residues as alpha-helical or non-alpha-helical. By leveraging data from Expasy, UniProt, and annotations of protein structures, we aimed to build and evaluate predictive models. This report details the steps taken, from data preprocessing to model evaluation, and highlights the Random Forest model's performance.

Data Exploratory Analysis

The data used for this study originated from the following three primary sources:

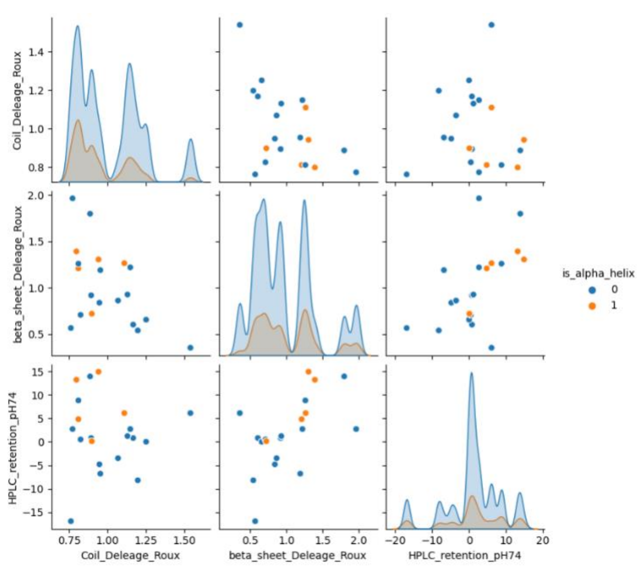
- Expasy_AA_Scales.txt
- UniProt2Seq.txt
- samples.txt.

These datasets provided biochemical scales, amino acid sequences, and structural labels, respectively. The data preprocessing phase involved merging these datasets into a unified dataset. Rows containing missing or infinite values were removed to ensure data integrity and quality.



Exploratory analysis revealed several key insights. The **correlation map** indicated that many biochemical scales were highly interrelated. For example, beta_sheet_Deleage_Roux showed a moderate correlation (0.57) with HPLC_retention_pH74, reflecting overlapping yet distinct biochemical roles. Features with correlation coefficients greater than 0.95 were removed to reduce redundancy and improve model interpretability, ensuring that the models could focus on truly independent signals.

Furthermore, the **pair plot** highlighted clear separations between alpha-helical (is_alpha_helix=1) and non-alpha-helical (is_alpha_helix=0) residues for features such as Coil_Deleage_Roux. Clusters observed in these visualizations reinforced the hypothesis that specific structural features significantly contribute to residue classification.



Overall, this analysis validated the relevance of key features and set the stage for effective feature engineering. By revealing patterns and correlations, it provided essential insights for the subsequent machine learning pipeline.

Feature Engineering

Feature engineering sought to amplify the model's ability to generalize across diverse protein contexts. While raw features such as polarity and hydrophobicity were foundational, engineered features provided contextual relevance, ensuring the model captured both local and global structural information.

Rolling Mean Features

Capturing local neighborhood effects was critical for understanding sequence context. For each residue, rolling averages over a ±2 residue window were computed for hydrophobicity, beta-sheet propensity, and HPLC retention. This smoothing approach blurred local variations and emphasized regional patterns critical for predicting secondary structures, especially alpha-helices.

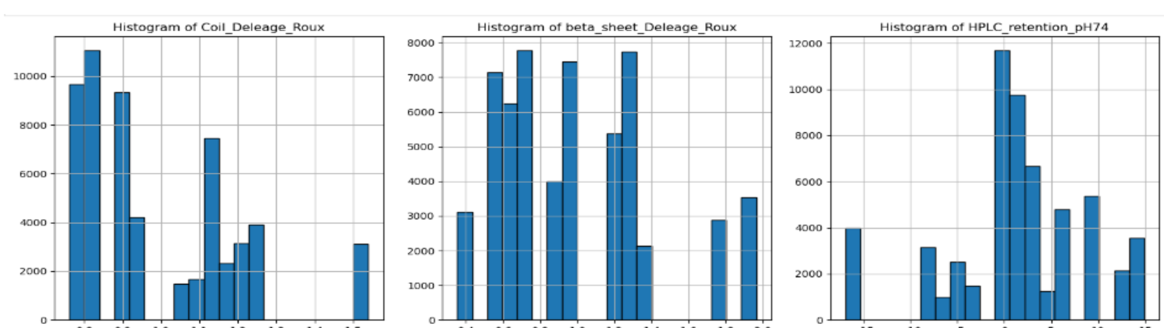
Ranking Features

Within individual proteins, ranking features by their relative magnitude provided new perspectives on residue significance. For instance, residues ranked high for coil propensity often correlated negatively with alpha-helical annotations. Ranking ensured invariance to global scaling differences across proteins, making the model robust to variability across datasets.

Feature Selection

Using the Random Forest model's feature importance scores, redundant and less informative features were pruned. Features with importance scores below 0.01 were excluded, reducing the feature space to 35 dimensions. This step streamlined computations without sacrificing predictive power. The importance scores underscored the role of hydrophobicity and conformational propensities in distinguishing alpha-helical residues.

The **histogram plots** below illustrate the transformed distributions for the key features after engineering. For example, hydrophobicity distributions for alpha-helical residues shifted towards higher values, while coil propensity exhibited a broader spread. These engineered features significantly improved the model's ability to differentiate structural classes, as reflected in subsequent model evaluation.



Results | Model Training & Hyperparameter Tuning

The training process began with splitting the dataset into distinct training and testing subsets, ensuring no overlap between proteins in the two sets to prevent data leakage and ensure robust evaluation. The training set was used to fit the models, while the test set provided an independent evaluation of performance. Three machine learning models were compared: K-Nearest Neighbors (KNN), Logistic Regression, and Random Forest. Each model underwent five-fold cross-validation to validate consistency and reliability, with accuracy as the primary metric for comparison.

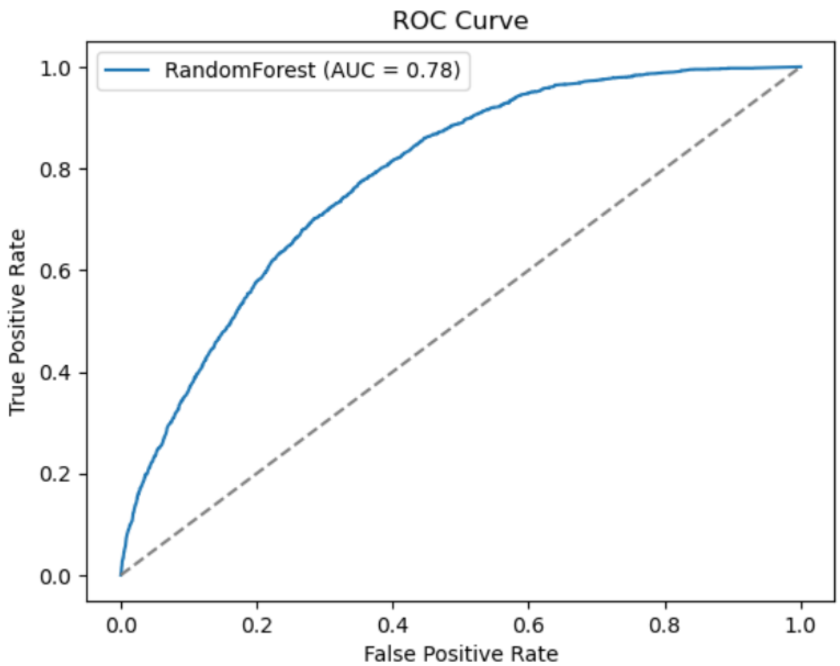
Among the three models, Random Forest emerged as the best performer, achieving a test set accuracy of 78.4%. Its ability to model nonlinear relationships and feature interactions contributed significantly to its success. In comparison, KNN achieved an accuracy of 77.55%, struggling with high-dimensional data, while Logistic Regression, being linear, attained 77.19% and lacked the capacity to model complex relationships.

Model Accuracies:

- KNN: *77.55%*
- Random Forest: *78.35%*
- Logistic Regression: *77.19%*

Hyperparameter tuning further refined the Random Forest model. A comprehensive grid search identified optimal values for the number of trees (**n_estimators = 200**), maximum tree depth (**max_depth = 10**), and class weighting (**class_weight = 'balanced'**). These settings improved generalization and mitigated the impact of class imbalance, ensuring more equitable treatment of the minority class (alpha-helical residues)

Best Model:



As seen from the Results section, The Random Forest model stood out as the top performer, reaching **78.4% accuracy** on the test set. As illustrated in the ROC curve below (AUC = 0.78), the model demonstrated strong discriminative ability in distinguishing alpha-helical residues from non-alpha-helical residues. Despite class imbalance leading to higher precision and recall for non-alpha-helical residues, the overall performance underscores Random Forest's robustness for secondary structure prediction.

The table below provides a detailed classification report. While the majority class (0: Non-alpha-helix) exhibits notably higher recall than the minority class (1: Alpha-helix), the weighted metrics remain strong. These results confirm the model's suitability for large-scale protein analysis, particularly when combined with further techniques—such as oversampling or adjusted loss functions—to address class imbalance.

Below are our results of our random forest model that had a 78.4% accuracy.

Classification Report on Test Set for Random Forest:

	precision	recall	f1-score	support
0	81.2%	93.6%	87.0%	8842
1	55.8%	27.1%	36.5%	2632
Macro Avg	68.5%	60.3%	61.7%	11474
Weighted Avg	075.4%	78.4%	75.4%	11474

Discussion

This study demonstrates the feasibility of using machine learning to predict protein secondary structures. By integrating biochemical scales and sequence context, the models successfully captured key patterns to classify residues effectively. These features highlighted both residue-specific and contextual properties, enabling a nuanced approach to the classification task.

However, certain limitations affected the study's outcomes. One of the primary challenges was the inherent class imbalance within the dataset, where non-alpha-helical residues dominated. This imbalance negatively impacted the recall for alpha-helical residues, leading to higher false negatives. Additionally, the binary classification approach, while simplifying the task, excluded critical secondary structures like beta-sheets and loops, which are important for understanding the complete protein folding landscape.

To address these limitations, future research should prioritize expanding the dataset to include more proteins and diverse structural elements. Increasing the representation of alpha-helical residues can directly tackle the issue of class imbalance. Incorporating evolutionary information, such as position-specific scoring matrices derived from sequence alignment, could enhance the model's understanding of functional and structural residue contexts.

Exploring deep learning architectures, such as convolutional and recurrent neural networks, is another promising direction. These models are adept at capturing complex patterns and long-range dependencies, potentially improving predictive accuracy for diverse secondary structures. Moreover, transfer learning approaches could leverage pre-trained protein models to boost performance on specific datasets.

Finally, integrating advanced visualization techniques and model interpretability tools can provide deeper insights into the model's decision-making process. This could guide biologists in understanding the relevance of specific features, paving the way for further exploration in computational biology and its applications.

Conclusion

This study highlights both the strengths and limitations of using machine learning to predict protein secondary structures, offering valuable insights and a foundation for further advancements. The Random Forest model showcased its ability to effectively classify residues, leveraging engineered features to capture both local and global properties of protein sequences. Its strengths include handling nonlinearity, interpretability of feature importance, and competitive performance relative to simpler models like Logistic Regression and K-Nearest Neighbors.

However, weaknesses such as class imbalance and the binary nature of the task underscore the need for refinement. Alternative approaches, such as Gradient Boosting, could be explored to further enhance performance by minimizing overfitting and increasing robustness. Additionally, advanced methods like ensemble learning or incorporating domain-specific knowledge could address limitations in predictive power for minority classes.

The implications of this work extend beyond secondary structure prediction. By providing a framework for feature engineering and model evaluation, this study sets the stage for tackling more complex protein structure problems, including tertiary and quaternary structures. Machine learning models could be combined with existing biological tools to improve accuracy and applicability in real-world scenarios, such as drug design or functional annotation of unknown proteins.

Incorporating deep learning techniques, particularly convolutional neural networks or recurrent architectures, holds promise for capturing intricate dependencies in protein sequences. Transfer learning, using pre-trained models on larger biological datasets, could also be a game-changer for smaller-scale studies like this one.

Overall, this study demonstrates the utility and adaptability of machine learning in bioinformatics. While the current Random Forest implementation provides a solid foundation, the exploration of advanced algorithms and richer datasets will undoubtedly pave the way for groundbreaking discoveries in protein research and computational biology.

Appendix

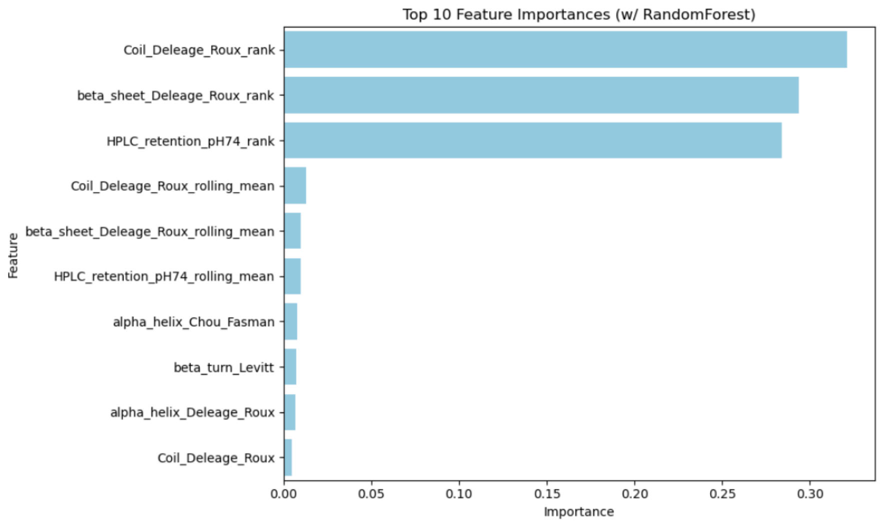


Figure 1- Random Forest Top Features

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